APPENDIX 6.6

Measurement of Capacity Utilization

The analysis of the rate of profit requires us to distinguish its structural trend from fluctuations arising from cycles and shocks. The key step here is to distinguish between economic capacity and economic output, the latter being linked to the former by the rate of capacity utilization.

It is important at the outset to distinguish between "engineering capacity," which is the maximum sustained production possible over some interval, and "economic capacity," which is the desired level of output from a given plant and equipment. For instance, it may be physically feasible to operate a plant for 20 hours per day 6 days a week, for a total of 120-hours per week of engineering capacity. But it may turn out that the potentially higher costs of second and third shifts make the lowest cost point consistent with only a single 8-hour shift per day for five days a week (i.e., 40-hours per week). This lowest cost point defines economic capacity, the firm's benchmark level of output, which in this example would represent a 33.3% rate of utilization of engineering capacity (chapter 4, section V). Production persistently below this level would signal the need for a slowdown in planned capacity growth, while that persistently above it would signal the need for accelerated capacity growth (Foss 1963, 25; Kurz 1986, 37–38, 43–44; Shapiro 1989, 184).[1] It is always the economically desired utilization rate which is the key.

Economic capacity is also different from "full employment output."[2] Since both measures have been labeled "potential output," it is important to distinguish them. Even though standard economic theory typically assumes that full capacity and full employment occur simultaneously, in actual practice there is no reason to suppose that production at economic capacity would serve to fully employ the existing labor force. Indeed, within classical, Keynesian, and Kaleckian traditions, the two are distinct even at the theoretical level (Garegnani 1979).

I. Conventional Measures of Capacity Utilization

It might be supposed that one could distinguish between actual output and capacity by identifying the latter as the long run "trend" of real output. But because aggregate data may contain asymmetric shocks, multiple cycles, and even long "waves," it becomes difficult to specify any

[1] In a growing system, adjustments take place by changes in relative growth rates.

[2] Short-run fluctuations in employment are likely to be correlated with short-run fluctuations in output. Thus, short-run fluctuations in the employment rate (employment over labor supply) are likely to be correlated with short-run fluctuation in the capacity utilization rate (output over capacity). But unless the ratio of capacity to labor supply, which is a kind of potential productivity of labor, happened to be always constant over time, the capacity utilization rate will deviate from the employment rate over the medium and long runs.

such trend. One procedure is to specify the trend as some a priori function of time. But there is little reason to believe that growth trends are independent of actual rates of growth, and there is no real reason to prefer one time function over any other. Another procedure is to smooth the data (as with the Hodrick–Prescott Filter) so as to bring out the trend. Here, one must have an a priori preference for the degree of "stiffness" to assign to the trend (e.g., the size of the HP Filter parameter). Moreover, some smoothing methods can give rise to spurious long cycles (Harvey and Jaeger 1993, 234). In either case, the chosen trend need not represent the path along which capacity utilization is normal, so that the residual fluctuations in the data may be a combination of variations in capacity utilization and those of the normal capacity path itself. Finally, there is the problem that fluctuations brought about by depressions, wars, and various other major conjunctural events are not generally symmetric. Smoothing techniques tend to split the data evenly between "ups and downs," which means they generally misrepresent the actual deviations from the trend. For instance, in the case of the Great Depression with its sharp collapse and protracted trough, the distortion would be quite significant. The oil price shocks of the 1970s would present similar difficulties.

An alternate approach is to try to estimate economic capacity directly. This would be relatively simple matter if one could accept the widely held (neoclassical) assumption that, except for downturns associated with the short (three- to five-year) cycle, capitalist economies generally operate at normal capacity. Indeed, this is the premise of the well-known Wharton method, which defines capacity as the peak output achieved in each business cycle or conjunctural fluctuation. The implicit assumption that all short-run peaks in output represent the same level (100%) of capacity utilization (Hertzberg, Jacobs, and Trevathan 1974; Schnader 1984) automatically excludes the possibility of medium- and long-term variations in capacity utilization.

A second group of capacity measures tries to get around this problem by relying on economic surveys of operating rates, as in those by the Bureau of Economic Analysis (BEA) and the Bureau of the Census. Here, firms are typically asked to indicate their current operating rate (i.e., their current rate of utilization of capacity). The difficulty with such surveys is that they do not specify any explicit definition of what is meant by "capacity," so that the respondents are free to choose between various measures of capacity, and the analysts who use this data are free to interpret them in manners consistent with their own theoretical premises. A typical case in point is the widely used Federal Reserve Board (FRB) measure of capacity utilization in manufacturing. It begins with a preliminary estimate of capacity by using two different surveys, one by McGraw-Hill (recently discontinued) and one by the Bureau of the Census. The Federal Reserve first combines them in some manner whose details it does not make public. Yet it frequently concludes that the resulting estimates of capacity utilization are not plausible even from their own point of view, so it further operates on the combined capacity measures to smooth them out, using regressions on the capital stock and on time (Shapiro 1989, 185–187). Various other adjustments are also made so as to "move the capacity estimate from a peak engineering concept toward an economic concept" consistent with its underlying theory. It is one of the stated goals of these adjustments that the resulting measure of capacity utilization rate is not "chronically below 'normal' capacity utilization" (Shapiro 1989, 187–188). *In other words, just as in the case of the Wharton method, the central premise here is that the economic system generally operates at, or near, full capacity*.

A third procedure, as employed by the IMF and the OECD, estimates potential output by means of fitted production functions. As has been often pointed out, a production function represents the optimal output which can be produced given fully utilized capital and labor inputs (Fisher 1969). Since actual capital and labor cannot be assumed to be fully utilized at any moment (this being the problem under investigation), this method requires some adjustment of the inputs. Thus, potential output is estimated using a labor input defined by the natural rate of unemployment and a capital input defined by the trend level of total factor productivity for that particular labor input (De Masi 1997). Needless to say, this requires theoretical faith not only in the much criticized notion of an aggregate production function (McCombie 2000–2001; Felipe and Fisher 2003; Shaikh 2005) but also in the existence of a natural rate of unemployment (see chapters 12–14 for a critical view).

A fourth type of measure sidesteps the difficulties inherent in the first two by attempting to directly measure the rate of capacity utilization. In a now classic study, Foss (1963) showed that it is possible to estimate capacity utilization by measuring the utilization rate of the electric motors used to drive capital equipment. Foss's initial estimates for selected years were subsequently developed into an annual series by Jorgenson and Grilliches (1967) and then improved and extended by Christensen and Jorgenson (1969) to cover the period from 1929 to 1967, and by Shaikh (1992) to cover the period from 1909 to 1928. But there exists a major obstacle to the forward extension of this series: namely, that the data on the installed capacity of electric motors, which is crucial to the construction of the series, was dropped after the 1963 Census. Shaikh (1987a) showed that direct survey data available from McGraw-Hill yielded a measure of capacity utilization that was very similar to that derived from data on electric motor use, in their periods of overlap. This allowed him to splice the two series together to create a complete capacity utilization series from 1947 to 1985 which differed significantly from the standard FRB measure (Shaikh 1987a), particularly in terms of longer run patterns such as the Vietnam War buildup during the 1960s, and post-Reagan profit boom from 1982 onward. Unlike the FRB measure, Shaikh's measure is neither symmetric nor stationary over the long run, and it exhibits much greater fluctuations. Conversely, the capacity–output ratio it yields has a much smoother trend than that derived from the FRB measure. Further detail is provided in Shaikh (Shaikh 1987a, 1992, 1999) and additional discussion can be found in Winston (1974), Gabish and Lorenz (1989, 26–40), and Tsaliki and Tsoulfidis (1999).

II. A New Approach to Measuring Capacity Utilization

The profit rate $r \equiv P/K$ can be written as the product of the profit share (σ_P) and the output–capital ratio (R, the observed maximum rate of profit).[3] The latter can in turn be expressed as the product of the capacity–output ratio ($R_n = Y_n/K$, the normal capacity maximum rate of profit) and the capacity utilization rate ($u_K = Y/Y_n$, the ratio of actual output to normal capacity output). The normal rate of profit is the product of the normal profit share and the capacity–capital ratio, while the actual rate of profit is the normal rate times the rate of capacity utilization (Shaikh 1999, 108).[4] As previously noted, r and R are inflation-adjusted measures when both

[3] This methodology was originally developed in Shaikh (2005).

[4] The output–capital ratio is a flow–stock ratio and can be directly expressed as the product of the capacity–capital ratio and the capacity utilization rate. Strictly speaking, we should also try to partition the profit share into the product of a normal (capacity-utilization-adjusted) component and a cyclical component. However since the profit share is a flow–flow ratio, changes in capacity utilization affect both sides so the relative fluctuation is much less. One could use the trend in the profit share as a proxy for its normal level (chapter 6, section VIII).

the numerator and denominator are expressed in the same units, that is, both in current cost, or both in constant cost converted through the same price index (see appendix 6.2). This is relevant when we make use of real output and real capital stock separately as in equations (6.6.3) and (6.6.4).

$$r = \left(\frac{P}{Y}\right)\left(\frac{Y}{K}\right) = \sigma_P \cdot R \tag{6.6.1}$$

$$R \equiv \left(\frac{Y}{K}\right) = \left(\frac{Y_n}{K}\right)\left(\frac{Y}{Y_n}\right) = R_n \cdot u_K \tag{6.6.2}$$

$$r_n \equiv \sigma_{Pn} \cdot R_n \tag{6.6.3}$$

Equation (6.6.2) for the output–capital ratio presents us with a simple means of estimating the capacity utilization rate if we expand it into its component parts displayed in equation (6.6.4). Given that output and capital stock are known, this becomes an unobserved component problem in two variables R_n, u_k. We can close the model through a technical progress function in which the capacity–capital ratio is considered to be a function of time (representing autonomous technical change) and a function of the capital stock (representing embodied technical change) subject to random shocks ε_t. The functional form depicted in equation (6.6.5) encompasses Harrod-neutral technical change ($b = c = 0$, so that R_n is stationary) and capital-biased technical change ($c < 0$ such that R_n falls as the capital stock grows) (Hahn and Matthews 1970, 371–372; Eltis 1984, 280–285; Foley and Michl 1999, 117–119). In growth terms, it allows for the possibility that the rate of change of the capacity–capital ratio may have an autonomous component and an embodied one which depends on the rate of capital accumulation.[5] Combining these yields a potential econometric relation.

$$\ln Y = \ln K + \ln R_n + \ln u_K \tag{6.6.4}$$

$$\ln R_n = a + b \cdot t + c' \cdot \ln K + \varepsilon_t \tag{6.6.5}$$

From a classical point of view, output gravitates around normal capacity so that $u_K \approx 1$ and $\ln u_K \approx 0$ over some long-run process. Economic capacity as defined here refers to normal capacity, not engineering capacity.

$$\ln Y \approx a + b \cdot t + c \cdot \ln K + \varepsilon_t \text{ where } c = 1 + c' \tag{6.6.6}$$

The intuitive idea is that economic capacity is that aspect of output which is cointegrated with the capital stock over the long run, subject to a trend in the capital–capacity ratio due to technical change. But now the issue of inflation becomes relevant. Rising prices will raise Y and K and impart a common component to both sides of equation (6.6.6) independently of any structural relation between them. It was previously established that the proper economic measure $R \equiv Y/K$ must be unit-free, so that its numerator and denominator both be measured in current prices or alternately both deflated by a common price index. Calculated in that manner, it is a real measure. The same issue crops up in equation (6.6.6): to get rid of the inflationary

[5] For any variable x the percentage rate of change is $d\ln x/dt = \dot{x}/x$. Hence, $\dot{R}_n/R_n = b + c\left(\dot{K}/K\right)$ from equation (6.6.5).

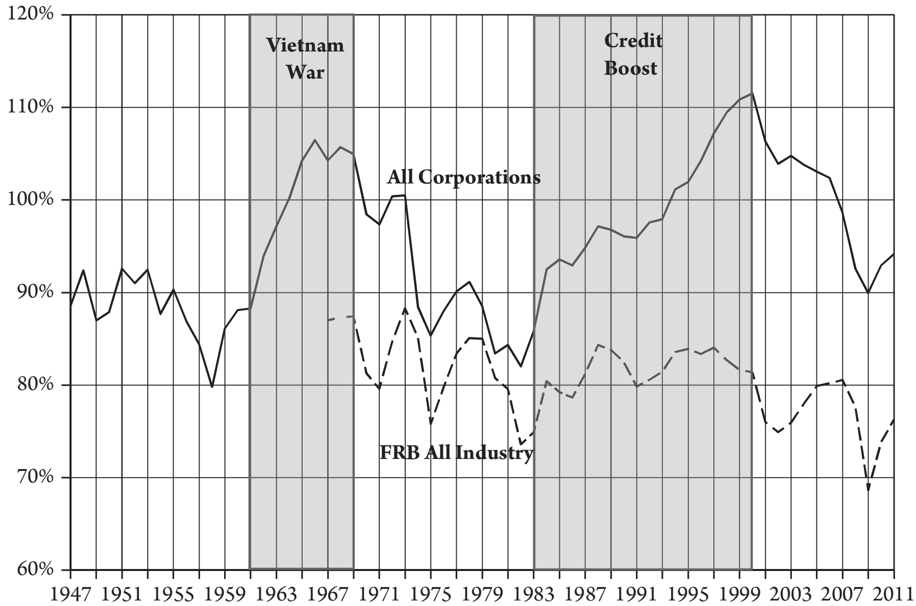
effect, we can only subtract the log of a common price index (p) from both sides (i.e., deflate Y and K by a common price p) because otherwise we would introduce a spurious relative price term into the equation (appendix 6.2, section III). Hence, the profit-rate-consistent method of estimating capacity utilization can be summarized as shown below.

$$\ln\left(\frac{Y}{p}\right) \approx a + b \cdot t + c \cdot \ln\left(\frac{K}{p}\right) + \varepsilon_t \tag{6.6.7}$$

Once output and capital stock have been restated in common price terms, the problem is easily addressed through time series methods, as delineated in data appendix 6.7, section III. The general method is easy to implement and requires only widely available data on real output and real capital stock provided both are deflated by the same price index. The cointegration model developed there allows us to estimate capacity and hence capacity utilization, the speed of adjustment of between output and capacity, and the direction of technical change. For instance, Harrod-neutral technical change implies a constant capacity–capital ratio, while capital-biased technical change implies a falling one (Michl 2002, 278). We will see that the postwar period in the United States is strongly characterized by capital-biased change. When applied to US manufacturing data this method also closely replicates a previously developed independent measure of capacity utilization constructed from census and survey data (Shaikh 1987a, appendix B). A further advantage of this procedure is that it does not require the assumption of an aggregate production function or some natural rate of unemployment as in the NAIRU or other hypotheses.

It should be said that the question of capacity utilization could be approached without regard to a consistent treatment of the maximum or actual rate of profit. From that point of view, the appropriate relation is between real output and real capital stock, each defined in terms of its own price index. But then one runs directly into the issue of quality adjustment discussed in appendix 6.5, section III. There are only two basic approaches to the construction of price indexes: (1) the traditional observed-price approach; and (2) the more recent quality-adjusted price approach. Their purposes, and hence their outcomes, are very different. The current-cost capital stock is simply the list of capital goods in the physical stock evaluated at current market prices. In the observed-price method, we deflate this by a price index of capital goods relative to their prices in some base year. The traditional measure of real capital is therefore simply the physical stock evaluated at base year prices. The more recent quality-adjusted price approach seeks to redefine prices so that they refer not to actual commodities but rather to the putative "user benefits" associated with these commodities. What we end up with is a measure of "quality" of the physical stock. Hence, the observed-price approach defines real capital as the constant dollar value of the stock of machines, while the quality-adjusted approach defines real capital as the constant dollar value of the "quality" embodied in these machines. The trouble is that within neoclassical theory the relevant definition of the "quality" of a capital good is its real profit: "the correct theoretical concept of capital is to consider two capital goods as equivalent if they generate the same [real gross profit]" (Gordon 1993, 103).

It follows that a perfectly implemented neoclassical quality-adjusted capital stock would yield a constant real rate of profit. Moreover, any variations in the real output–capital ratio would then reflect those in the profit share: properly implemented quality-adjusted price indexes of capital goods would suppress any trend due to technical change and the measure of capacity utilization would be correspondingly distorted. In practice quality adjustment is not perfect, so some



Appendix Figure 6.6.1 Capacity Utilization, US Corporation 1947–2011 and FRB All Industry

aspect of technical change may appear. But to read these results as valid indicators of technical change and capacity utilization would be erroneous, since they are artifacts of quality adjustment.

One of the virtues of the profit-rate-consistent method of estimating capacity utilization developed in this book is that it avoids these problems. Appendix figure 6.6.1, which is the same as figure 6.4 at the end of chapter 6, displays the measure of capacity utilization of US corporations, which is quite different in trend from the conventional FRB measure of industrial capacity utilization (appendix 6.7, section III). Note the great impact of the Vietnam War boom in the Johnson era of the 1960s, and the corresponding credit-fueled demand boom of the Reagan–Bush Sr. era of the 1980s–1990s in the new measure. Such demand-pumping episodes turn out to play important theoretical and empirical roles in chapters 12–16.